Custom CAN Communication Protocol_V3.07b0

Protocol Versions

- Rev.3.01b0 – 01.06.2024: Initial version.
- Rev.3.02b0 – 22.07.2024: Optimized system reset protocol.
- Rev.3.03b0 – 01.08.2024: Added PID parameter configuration commands for position and speed closed-loop control.
- Rev.3.04b0 – 21.11.2024: Added brake output control; added encoder fault to fault code list.
- Rev.3.05b0 – 18.03.2025:
  - Corrected descriptive errors in the document.
  - The slave can receive and respond to commands whose ID is (0x100 | Dev_addr).
  - When a fault is detected, the slave actively reports the status information corresponding to command 0xAE.
  - Improved communication examples (see the end of this document).
- Rev.3.05b1 – 03.04.2025: Optimized document content and corrected incorrect descriptions.
- Rev.3.06b0 – 23.04.2025: Added MIT-type motion-control protocol mode.
- Rev.3.07b0 – 18.05.2025:
  - Supplemented the description of the “Set current position as home” command.
  - Added command 0xA4.

1. CAN Interface and Addressing

The default CAN baud rate is 1 MHz. It can be configured via the host PC to 1 MHz, 500 kHz, 250 kHz, 125 kHz or 100 kHz.
All CAN messages in this protocol use data frames with standard identifiers (11-bit StdID), and all multi-byte values use little-endian byte order.

When shipped, the default device address (Dev_addr) is 0x01. It can be modified by the host PC to any value in the range 1…254.

Two special addresses are defined by the system:
- 0x00 – Broadcast address. All slaves execute the command but do not reply to the master.
- 255 (0xFF) – Public address. All slaves will respond to commands addressed to 0xFF, and their reply ID will be their own Dev_addr.

To distinguish whether a data frame on the bus is sent by the master or is a reply from a slave, the following mechanism is used.
Besides responding to commands whose StdID equals its own Dev_addr, a slave can also recognize and respond to commands whose StdID is (0x100 | Dev_addr). In either case, the reply frame ID from the slave to the master is always Dev_addr.

Example: if the slave’s Dev_addr is 0x01 and the master sends a data frame with StdID = 0x101, the slave’s reply will have StdID = 0x01. In this way the communication direction on the CAN bus can be determined from the frame ID.

The current device address can also be identified from the blinking pattern of the green LED on the driver board:

• Address 1: [__ -__-__]
• Address 2: [__-_-__-_-__]
• Address 3: [__-_-_-__-_-_-__]

2. Data Type Description

• 1u – 1 unsigned byte
• 1s – 1 signed byte
• 2u – 2 unsigned bytes (little-endian)
• 2s – 2 signed bytes (little-endian)
• 4u – 4 unsigned bytes (little-endian)
• 4s – 4 signed bytes (little-endian)
• 4f – 4-byte single-precision floating-point value (little-endian)
• b – Number of bytes
• bit – Bit; 1 byte = 8 bits

3. Supported Custom CAN Control Commands

Category	Command	Description
System		
	0x00	Reset slave. After receiving this command from the master, the slave immediately resets and does NOT reply.
	0xA0	Read Boot, application software, hardware, and custom CAN protocol versions.
	0xA1	Read real-time Q-axis current.
	0xA2	Read real-time rotational speed.
	0xA3	Read real-time single-turn absolute angle and multi-turn absolute angle.
	0xA4	Read real-time temperature, Q-axis current, speed, single-turn absolute angle.
	0xAE	Read real-time bus voltage, bus current, working temperature, run mode and fault-code status. Once the slave detects a fault, it reports real-time status every 200 ms using this command.
	0xAF	Clear faults.
Parameter		
	0xB0	Read motor pole pairs, torque constant and gear ratio.
	0xB1	Set current position as home. (Single-turn absolute zero will be stored in the driver and retained after power-off.)
	0xB2	Set maximum speed for position mode (not saved after power-off).
	0xB3	Set maximum Q-axis current for position or speed mode (not saved after power-off).
	0xB4	Set Q-axis current ramp rate in current-control mode (not saved after power-off).
	0xB5	Set acceleration in speed-control mode (not saved after power-off).
	0xB6	Read or set position-loop Kp (not saved after power-off).
	0xB7	Read or set position-loop Ki (not saved after power-off).
	0xB8	Read or set speed-loop Kp (not saved after power-off).
	0xB9	Read or set speed-loop Ki (not saved after power-off).

Control
 0xC0 Q-axis current control. (Torque = torque_constant × Q-axis current.)
 0xC1 Speed control.
 0xC2 Absolute position control.
 0xC3 Relative position control.
 0xC4 Move motor back to the defined home position along the shortest path, with a travel angle not exceeding 180°.
 0xCE Brake (holding brake) output on/off control.
 0xCF Disable motor output. Motor enters free state without control (this is the default state after power-on).

MIT-type Motion-Control Protocol
 0xF0 Read and configure Pos_Max, Vel_Max and T_Max used in the motion-control mode.
 0xF1 Read real-time position, speed, torque and status in motion-control mode.
 –   Motion-control command frame without command code; the highest bit (Bit[10]) of StdID is set to 1.

4. Command Details

4.1 Command 0x00 – Reset Slave

Master → Slave
• StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
• DLC: 0x08
• Data[0]: 0x00 (command code)
• Data[1]–Data[7]: 0xFF 0x00 0xFF 0x00 0xFF 0x00 0xFF (padding data)

Behaviour: After receiving this command, the slave immediately restarts and does not reply to the master.

4.2 Command 0xA0 – Read Boot / Software / Hardware / Custom CAN Protocol Version

Master → Slave
• StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
• DLC: 0x01
• Data[0]: 0xA0

Slave → Master
• StdID: Dev_addr
• DLC: 0x08
• Data[0]: 0xA0 (command code)
• Data[1]–Data[2]: Boot firmware version (2u)
• Data[3]–Data[4]: Application firmware version (2u)
• Data[5]–Data[6]: Hardware version (2u) – driver board hardware ID
• Data[7]: Custom CAN protocol version (1u)

4.3 Command 0xA1 – Read Real-time Q-axis Current

Master → Slave
• StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
• DLC: 0x01
• Data[0]: 0xA1

Slave → Master
• StdID: Dev_addr
• DLC: 0x05
• Data[0]: 0xA1
• Data[1]–Data[4]: Q-axis current (4s), unit 0.001 A
  Torque = torque_constant × Q-axis current.

4.4 Command 0xA2 – Read Real-time Speed

Master → Slave
• StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
• DLC: 0x01
• Data[0]: 0xA2

Slave → Master
• StdID: Dev_addr
• DLC: 0x05
• Data[0]: 0xA2
• Data[1]–Data[4]: Speed (4s), unit 0.01 rpm

4.5 Command 0xA3 – Read Real-time Single-turn and Multi-turn Absolute Angles

Master → Slave
• StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
• DLC: 0x01
• Data[0]: 0xA3

Slave → Master
• StdID: Dev_addr
• DLC: 0x07
• Data[0]: 0xA3
• Data[1]–Data[2]: Single-turn absolute angle (2u)
  Angle° = value × (360 / 16384).
• Data[3]–Data[6]: Multi-turn absolute angle (4s)
  TotalAngle° = value × (360 / 16384).

4.6 Command 0xA4 – Read Real-time Temperature, Q-axis Current, Speed and Single-turn Angle

Master → Slave
• StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
• DLC: 0x01
• Data[0]: 0xA4

Slave → Master
• StdID: Dev_addr
• DLC: 0x08
• Data[0]: 0xA4
• Data[1]: Working temperature (1u), unit: °C
• Data[2]–Data[3]: Q-axis current (2s), unit 0.001 A
• Data[4]–Data[5]: Speed (2s), unit 0.01 rpm
• Data[6]–Data[7]: Single-turn absolute angle (2u)
  Angle° = value × (360 / 16384).

4.7 Command 0xAE – Read Bus Voltage, Bus Current, Temperature, Run Mode and Fault Codes

Once the slave detects a fault, it will automatically send this status frame every 200 ms.

Master → Slave
• StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
• DLC: 0x01
• Data[0]: 0xAE

Slave → Master
• StdID: Dev_addr
• DLC: 0x08
• Data[0]: 0xAE
• Data[1]–Data[2]: Bus voltage (2u), unit 0.01 V
• Data[3]–Data[4]: Bus current (2u), unit 0.01 A
• Data[5]: Working temperature (1u), unit: °C
• Data[6]: Run mode (1u)
  0 – Disabled
  1 – Voltage control
  2 – Q-axis current control
  3 – Speed control
  4 – Position control
• Data[7]: Fault code (1u)
  Bit0 – Voltage fault
  Bit1 – Current fault
  Bit2 – Temperature fault
  Bit3 – Encoder fault
  Bit6 – Hardware fault
  Bit7 – Software fault
  Bitn represents bit n in the byte; Bit0 is the LSB.

4.8 Command 0xAF – Clear Faults

Master → Slave
• StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
• DLC: 0x01
• Data[0]: 0xAF

Slave → Master
• StdID: Dev_addr
• DLC: 0x02
• Data[0]: 0xAF
• Data[1]: Current fault code (1u) – bit definitions same as command 0xAE.

4.9 Command 0xB0 – Read Motor Pole Pairs, Torque Constant and Gear Ratio

Master → Slave
• StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
• DLC: 0x01
• Data[0]: 0xB0

Slave → Master
• StdID: Dev_addr
• DLC: 0x07
• Data[0]: 0xB0
• Data[1]: Motor pole pairs (1u)
• Data[2]–Data[5]: Torque constant (4f), unit N/A
• Data[6]: Gear ratio (1u)

4.10 Command 0xB1 – Set Current Position as Home

The current electrical/mechanical zero position will be stored in the driver board and retained after power-off.
If a second encoder is enabled in the system, execution time becomes longer and the response delay from the slave to the master
is about 35 ms.

Master → Slave
• StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
• DLC: 0x01
• Data[0]: 0xB1

Slave → Master
• StdID: Dev_addr
• DLC: 0x03
• Data[0]: 0xB1
• Data[1]–Data[2]: Mechanical angle offset (2u)

4.11 Command 0xB2 – Set Maximum Speed in Position Mode (Not Saved After Power-off)

Master → Slave
• StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
• DLC: 0x05
• Data[0]: 0xB2
• Data[1]–Data[4]: Position-loop output limit (4u), i.e. maximum speed in position mode, unit

0.01 rpm.

Slave → Master
• StdID: Dev_addr
• DLC: 0x05
• Data[0]: 0xB2
• Data[1]–Data[4]: Same as above.

4.12 Command 0xB3 – Set Maximum Q-axis Current in Position/Speed Mode (Not Saved After Power-off)

Master → Slave
• StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
• DLC: 0x05
• Data[0]: 0xB3
• Data[1]–Data[4]: Speed-loop output limit (4u), i.e. maximum Q-axis current in speed/position modes, unit 0.001 A.

Slave → Master
• StdID: Dev_addr
• DLC: 0x05
• Data[0]: 0xB3
• Data[1]–Data[4]: Same as above.

4.13 Command 0xB4 – Set Q-axis Current Ramp Rate in Current-Control Mode (Not Saved After Power-off)

Master → Slave
• StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
• DLC: 0x05
• Data[0]: 0xB4
• Data[1]–Data[4]: Q-axis current slew rate (4u), unit 0.001 A/s.

Slave → Master
• StdID: Dev_addr
• DLC: 0x05
• Data[0]: 0xB4
• Data[1]–Data[4]: Same as above.

4.14 Command 0xB5 – Set Acceleration in Speed-control Mode (Not Saved After Power-off)

Master → Slave
• StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
• DLC: 0x05
• Data[0]: 0xB5
• Data[1]–Data[4]: Acceleration (4u), unit 0.01 rpm/s.

Slave → Master

- StdID: Dev_addr
- DLC: 0x05
- Data[0]: 0xB5
- Data[1]–Data[4]: Same as above.

4.15 Commands 0xB6 / 0xB7 / 0xB8 / 0xB9 – PID Parameters (Position and Speed Loops)

For each of these commands, the usage is similar.

Command 0xB6 – Position-loop Kp
Command 0xB7 – Position-loop Ki
Command 0xB8 – Speed-loop Kp
Command 0xB9 – Speed-loop Ki

Master → Slave
- StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
- DLC: 0x01 to read, or 0x05 to write
- Data[0]: Command code (0xB6 / 0xB7 / 0xB8 / 0xB9)
- Data[1]–Data[4]: 4f parameter value (when writing). Omitted when DLC = 0x01.

Slave → Master
- StdID: Dev_addr
- DLC: 0x05
- Data[0]: Command code
- Data[1]–Data[4]: 4f parameter value.

4.16 Command 0xC0 – Q-axis Current Control

Master → Slave
- StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
- DLC: 0x05
- Data[0]: 0xC0
- Data[1]–Data[4]: Q-axis current (4s), unit 0.001 A
  Torque = Q_current × torque_constant.

Slave → Master
The reply contents are identical to command 0xA1, except that Data[0] is 0xC0.

4.17 Command 0xC1 – Speed Control

Master → Slave
- StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
- DLC: 0x05
- Data[0]: 0xC1
- Data[1]–Data[4]: Target speed (4s), unit 0.01 rpm.

Slave → Master
Reply format is the same as for command 0xA2, but with command code 0xC1.

4.18 Command 0xC2 – Absolute Position Control

Master → Slave
- StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
- DLC: 0x05
- Data[0]: 0xC2
- Data[1]–Data[4]: Absolute position (4s), unit: counts; one revolution = 16384 counts.

Slave → Master
Reply format is the same as for command 0xA3, but with command code 0xC2.

4.19 Command 0xC3 – Relative Position Control

Master → Slave
- StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
- DLC: 0x05
- Data[0]: 0xC3
- Data[1]–Data[4]: Relative position (4s), unit: counts; one revolution = 16384 counts.

Slave → Master
Reply format is the same as for command 0xA3, but with command code 0xC3.

4.20 Command 0xC4 – Return to Home Along Shortest Path

Master → Slave
- StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
- DLC: 0x01
- Data[0]: 0xC4

Slave → Master
Reply format is the same as for command 0xA3, but with command code 0xC4.

4.21 Command 0xCE – Brake Output Control

Master → Slave
- StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
- DLC: 0x02
- Data[0]: 0xCE
- Data[1]: Operation type
  0x00 – Brake output off (open)
  0x01 – Brake output on (closed)
  0xFF – Read current brake state

Slave → Master
- StdID: Dev_addr
- DLC: 0x02
- Data[0]: 0xCE

- Data[1]: Brake switch state
  0x00 – Open; 0x01 – Closed.

4.22 Command 0xCF – Disable Motor Output (Free State)

After power-on, the motor is in this disabled/free state by default.

Master → Slave
- StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
- DLC: 0x01
- Data[0]: 0xCF

Slave → Master
Reply format is the same as for command 0xAE, but with command code 0xCF.

5. MIT-type Motion-control Mode

To support different motor specifications, this protocol allows configuring the maximum position, speed and torque
according to the actual motor parameters and application requirements.

Note that in motion-control mode the units differ from those used earlier in this document:

1. Position unit: radian (rad), where 2π rad = 360°.
2. Speed unit: rad/s, where 2π rad/s = 60 rpm.
3. Torque unit: newton-metre (Nm). The correct value depends on the torque constant configured via the host PC.

A logical block diagram of the motion-control mode is shown on the original page.

5.1 Command 0xF0 – Read/Configure Pos_Max, Vel_Max and T_Max

Data is little-endian. The configured parameters are stored in the driver and are retained after power-off.

Master → Slave
- StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
- DLC: 0x01 to read, or 0x07 to configure
- Data[0]: 0xF0
- Data[1]–Data[2]: Pos_Max (2u), position limit in motion-control mode, unit 0.1 rad
- Data[3]–Data[4]: Vel_Max (2u), speed limit in motion-control mode, unit 0.01 rad/s
- Data[5]–Data[6]: T_Max (2u), torque limit in motion-control mode, unit 0.01 Nm

Slave → Master
- StdID: Dev_addr
- DLC: 0x07
- Data[0]: 0xF0
- Data[1]–Data[2]: Pos_Max (2u), default 955 → 95.5 rad

• Data[3]–Data[4]: Vel_Max (2u), default 4500 → 45.00 rad/s
• Data[5]–Data[6]: T_Max (2u), default 1800 → 18.00 Nm

5.2 Command 0xF1 – Read Real-time Position, Speed, Torque and Status in Motion-control Mode

Master → Slave
• StdID: Dev_addr, (0x100 | Dev_addr), 0x00 or 0xFF
• DLC: 0x01
• Data[0]: 0xF1

Slave → Master
• StdID: Dev_addr
• DLC: 0x07
• Data[0]: 0xF1
• Data[1]–Data[2]: Mechanical position (16 bits)
  – Byte[1] is high 8 bits, Byte[2] is low 8 bits
  – Range [0…65535] corresponds to [−Pos_Max … +Pos_Max]
• Data[3]–upper nibble of Data[4]: Mechanical speed (12 bits)
  – Byte[3] is high 8 bits, Byte[4] bits[7:4] are low 4 bits
  – Range [0…4095] corresponds to [−Vel_Max … +Vel_Max]
• Lower nibble of Data[4] and Data[5]: Torque (12 bits)
  – Byte[4] bits[3:0] are high 4 bits, Byte[5] is low 8 bits
  – Range [0…4095] corresponds to [−T_Max … +T_Max]
• Data[6]: Status (1u)
  – Bit0 = 1: Motion-control mode active
  – Bit1 = 1: System fault present

5.3 Motion-control Command Frame (No Command Code)

Because the host's command frame does not contain a command byte, the slave distinguishes motion-control frames by the
StdID: the highest bit Bit[10] of the 11-bit standard identifier must be set to 1.

Examples (Dev_addr = 1):
• StdID = 0x401 or 0x501, etc.

When the slave receives such a frame it immediately switches to motion-control mode and executes the command. To exit
motion-control mode, send command 0xCF. To set the current position as home, send command 0xB1.

Master → Slave
• StdID: (0x400 | Dev_addr), (0x400 | (0x100 | Dev_addr)), (0x400 | 0x00) or (0x400 | 0xFF)
• DLC: 0x08
• Data[0]–Data[1]: Target position, 16 bits
  – Data[0] high 8 bits, Data[1] low 8 bits
  – Range [0…65535] corresponds to [−Pos_Max … +Pos_Max]
• Data[2]–upper nibble of Data[3]: Target speed, 12 bits
  – Data[2] high 8 bits, Data[3] bits[7:4] low 4 bits

– Range [0…4095] corresponds to [−Vel_Max … +Vel_Max]
• Lower nibble of Data[3] and Data[4]: Position gain Kp, 12 bits
  – Data[3] bits[3:0] high 4 bits, Data[4] low 8 bits
  – Range [0…4095] corresponds to [0…500]
• Data[5]–upper nibble of Data[6]: Speed gain Kd, 12 bits
  – Data[5] high 8 bits, Data[6] bits[7:4] low 4 bits
  – Range [0…4095] corresponds to [0…5]
• Lower nibble of Data[6] and Data[7]: Target torque, 12 bits
  – Data[6] bits[3:0] high 4 bits, Data[7] low 8 bits
  – Range [0…4095] corresponds to [−T_Max … +T_Max]

Slave → Master
Reply format is the same as command 0xF1.

6. Example CAN Command Frames

All examples below assume little-endian byte order and standard data frames.

6.1 Command 0xAF – Clear Faults

Master sends (hex), DLC = 1:
  AF

Slave replies (hex), DLC = 2:
  AF 00

Interpretation of reply:
• 0x00 – Fault code: 0x00 means no fault; other values follow the fault-code definition in command 0xAF / 0xAE.

6.2 Command 0xC0 – Torque Mode, Target Q-axis Current = 1 A

Master sends (hex), DLC = 5:
  C0 E8 03 00 00
Here 0x000003E8 = 1000 decimal, 1000 × 0.001 = 1 A.

Slave replies (hex), DLC = 5:
  C0 F5 03 00 00
Interpretation:
• 0x000003F5 = 1013 decimal → current = 1013 × 0.001 = 1.013 A.

6.3 Command 0xC1 – Speed Mode, Target Speed = 100 rpm

Master sends (hex), DLC = 5:
  C1 10 27 00 00
Here 0x00002710 = 10000 decimal → 10000 × 0.01 = 100 rpm.

Slave replies (hex), DLC = 5:

```
  C1 38 27 00 00
```

Interpretation:
• 0x00002738 = 10040 decimal → speed = 10040 × 0.01 = 100.4 rpm.

6.4 Command 0xC2 – Absolute-position Mode, Target = 16384 Counts (1 Revolution)

Master sends (hex), DLC = 5:

```
  C2 00 40 00 00
```

Here 0x00004000 = 16384 decimal → motor rotates to 360° mechanical position.

Slave replies (hex), DLC = 7:

```
  C2 00 00 00 00 00 00
```

Interpretation of reply:
• Data[1]–Data[2] = 0x0000: current single-turn absolute position → 0°.
• Data[3]–Data[6] = 0x00000000: current multi-turn absolute position → 0° total.

6.5 Command 0xC3 – Relative-position Mode, Rotate +90° from Current Position

Master sends (hex), DLC = 5:

```
  C3 00 10 00 00
```

Here 0x00001000 = 4096 decimal; 4096 × (360/16384) = 90°.

Slave replies (hex), DLC = 7:

```
  C3 00 40 00 40 00 00
```

Interpretation of reply:
• Data[1]–Data[2] = 0x4000 (16384 decimal) → single-turn absolute position = 16384 × (360/16384) = 360°.
• Data[3]–Data[6] = 0x00004000 (16384 decimal) → multi-turn absolute position = 360°.

6.6 Command 0xCF – Disable Motor

Master sends (hex), DLC = 1:

```
  CF
```

Slave replies (hex), DLC = 8:

```
  CF 7C 09 01 00 26 00 00
```

Interpretation of reply fields:
• 0x097C = 2428 decimal → bus voltage = 2428 × 0.01 = 24.28 V.
• 0x0001 = 1 decimal → bus current = 1 × 0.01 = 0.01 A.
• 0x26 = 38 decimal → motor temperature = 38 °C.
• 0x00 – Run mode: 0 = disabled (see run-mode definition above).
• 0x00 – Fault code: 0 = no fault (see fault-code definition above).